Singularities in Different Characteristics

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1. Lecture–1 May 18, 2026

Singularities and Frobenius

Let $(R, \mathfrak{m})$ be a Noetherian local domain. We want to study the singularities of R.

Consider $R \hookrightarrow R'$ or $X \to \operatorname{Spec} R$. Another variant: Let A be a regular local ring such that $A \twoheadrightarrow R = A/I$. We want to understand I in terms of $A \to A'$.

Assume the characteristic is $p > 0$. Consider the Frobenius endomorphism:

$$\begin{aligned} F : R &\to R \\ r &\mapsto r^p \end{aligned}$$

One solution to the problem that the source and target of F are the same is to consider the roots. Let

$$R^{1/p} = \{x \in \overline{K(R)} \mid x^p \in R\},$$

which is the set of p-th roots of elements of R. We have the following diagram connecting the Frobenius map on R with the p-th power map on $R^{1/p}$:

$$\begin{tikzcd} R \arrow[r, "F"] \arrow[d, "\simeq"'] & R \arrow[d, "="] \\ R^{1/p} \arrow[r, "(\cdot)^p"'] & R \end{tikzcd}$$

Theorem 1.1 (Kunz). *Let $(R, \mathfrak{m})$ be a Noetherian local ring of characteristic $p > 0$. Then R is regular if and only if $R^{1/p}$ is a flat R-module. (Note that as rings, $R \simeq R^{1/p}$).*

Example 1.2. Let $R = k[x_1, \dots, x_n]_{\mathfrak{m}}$. Then $R^{1/p} = k^{1/p}[x_1^{1/p}, \dots, x_n^{1/p}]$. In this case, $R^{1/p}$ is free over R, and it has a basis:

$$1, x_1^{1/p}, \dots, x_n^{1/p}, x_1^{1/p}x_2^{1/p}, \dots, (x_1 \dots x_n)^{\frac{p-1}{p}}$$

More precisely, the basis is given by:

$$\left\{x_1^{a_1/p} \dots x_n^{a_n/p} \mid 0 \le a_i \le p-1\right\}$$

If $[k^{1/p} : k] = \infty$, then $R^{1/p}$ is flat but not free.

Proofs of Kunz's Theorem (Sketches)

Proof of Kunz's Theorem #1: We can reduce to the case where R is complete. Then $R \cong k[[x_1, \dots, x_n]]$ (or some variant of R does this via the Cohen Structure Theorem).

Proof of Kunz's Theorem #2: ($\Rightarrow$) Assume $R^{1/p}$ is a finite R-module (i.e., R is F-finite). By the Auslander-Buchsbaum formula (A-B),

$$\mathrm{pdim}_R(R^{1/p}) + \mathrm{depth}_R(R^{1/p}) = \mathrm{depth}(R) = \dim(R) \tag{*}$$

(Recall that regular implies Cohen-Macaulay (CM), so depth equals dimension).

Say $(x_1, \ldots, x_d) = \mathfrak{m} \subseteq R$, where $d = \dim R$. Since $x_1, \ldots, x_d$ is a regular sequence on R, it follows that $x_1^p, \ldots, x_d^p$ is a regular sequence on R. Under the isomorphism $R \xrightarrow{\cong} R^{1/p}$, the elements x_i form a regular sequence on $R^{1/p}$. This implies that $\mathrm{depth}_R(R^{1/p}) = \dim(R)$. By equation (*), we get $\mathrm{pdim}_R(R^{1/p}) = 0$. Thus, $R^{1/p}$ is projective, which implies it is flat.

Proof of Kunz's Theorem #3: Consider $R \to R^{1/p}$ as a local map of Noetherian rings such that for the fibers:

(1) R is regular.

(2) $\dim(R^{1/p}) = \dim R + \dim(R^{1/p}/\mathfrak{m}R^{1/p})$.

(3) $R^{1/p}$ is CM.

$\implies$ $R^{1/p}$ is flat over R.

Remark 1.3 (Explanatory Note)**.** The condition $\mathrm{pdim}(R^{1/p}) = 0$ implies the module is projective. Over a local ring, any finitely generated projective module is free, and free modules are flat.

F-split and Strongly F-regular Rings

Assume $R^{1/p}$ is a free R-module with some basis. If there exists a projection onto a factor, ρ, which is surjective, then $\rho(c^{1/p}) = 1$ for some c. Consider the mappings:

$$\begin{array}{ccccc} R & \xrightarrow{\cdot c^{1/p}} & R^{1/p} & \xrightarrow{\rho} & R \\ 1 & \longmapsto & c^{1/p} & \longmapsto & 1 \end{array}$$

Hence, $R \to R^{1/p}$ splits as a map of R-modules.

Definition 1.4. R is **F-split** if the natural inclusion $R \to R^{1/p}$ splits as a map of R-modules.

Now, assume R is regular and F-finite. Suppose $c \in \mathfrak{m}$, $c \neq 0$. Then $c \notin \mathfrak{m}^\ell$ for $\ell \gg 0$. So $c \notin (x_1^\ell, \ldots, x_d^\ell)$ (where $(x_1, \ldots, x_d) = \mathfrak{m}$). $\implies$ $c \notin (x_1^{p^e}, \ldots, x_d^{p^e})$ for $e \gg 0$.

Claim 1.5. *$c^{1/p^e} \in R^{1/p^e}/\mathfrak{m}R^{1/p^e}$ is non-zero.*

Notice the correspondence $\mathfrak{m}R^{1/p^e} \overset{\sim}{\leftrightarrow} (x_1^{p^e}, \ldots, x_d^{p^e}) \subseteq R$. So c^{1/p^e} is part of a basis for $R^{1/p^e}/\mathfrak{m}R^{1/p^e}$ over $R/\mathfrak{m}$. By Nakayama's Lemma (NAK) (using the F-finite assumption), c^{1/p^e} is a part of a basis for R^{1/p^e} over R. $\implies$ $\exists \rho : R^{1/p^e} \to R$ projecting onto c^{1/p^e} (i.e., mapping it to 1).

Definition 1.6. A ring R (which is F-finite) is **strongly F-regular (SFR)**, if $\forall c \neq 0$, the map $R \xrightarrow{\cdot c^{1/p^e}} R^{1/p^e}$ splits (as a map of R-modules) for some $e > 0$.

Lemma 1.7. *If R is F-finite and regular, then R is strongly F-regular (SFR).*

Remark 1.8. If you mod out a regular ring by an ideal $I \subseteq \mathfrak{m}^2$, the resulting quotient is never regular.

Example 1.9. 1. $\mathbb{F}_p[x, y, z]_{\mathfrak{m}}/(x^2 - yz)$ is NOT regular, but it is strongly F-regular, hence F-split.

Remark 1.10 (Explanatory Note)**.** This is an A_1 singularity. Note the standard isomorphism with the Veronese subring:

$$\frac{k[x,y,z]}{(x^2 - yz)} \simeq k[ab, a^2, b^2] \subseteq k[a,b]$$

2. $\dfrac{k[x,y,z]}{(x^3 + y^3 + z^3)}$ for $p \neq 3$ is NEVER strongly F-regular.

   (i) It is F-split if $p \equiv 1 \pmod 3$.

   (ii) It is not F-split if $p \equiv 2 \pmod 3$.

3. $\mathbb{F}_p[x, y, z]_{\mathfrak{m}}/(x^4 + y^4 + z^4)$ is NEVER F-split (even for $p \neq 2$).

How do you check these properties?

- **To show F-splitting:** Exhibit a splitting. If $R \hookrightarrow S$ splits and S is F-split, then R is F-split.

2. Lecture–2 May 19, 2026

Throughout this lecture, let R be a domain of finite type over a field k of characteristic 0. Frequently R is assumed to be local, say $(R, \mathfrak{m})$. We write

$$X = \operatorname{Spec} R.$$

What Is Resolution of Singularities?

Definition 2.1 (Resolution of singularities)**.** A *resolution of singularities* of $X = \operatorname{Spec} R$ is a proper, usually projective, birational morphism

$$\pi : Y \longrightarrow \operatorname{Spec} R$$

such that Y is non-singular.

In the affine setting, a typical projective birational morphism is obtained as a blowup. Namely, if $I \subseteq R$ is an ideal, then the blowup of $\operatorname{Spec} R$ along I is

$$Y = \operatorname{Proj} S, \qquad S = R \oplus It \oplus I^2t^2 \oplus \cdots .$$

Here $S = \bigoplus_{n \geq 0} I^n t^n$ is the Rees algebra of I. The adjective *birational* means that π is an isomorphism over a dense open subset $U \subseteq \operatorname{Spec} R$. Equivalently, Y and $\operatorname{Spec} R$ have the same function field:

$$k(Y) = \operatorname{Frac}(R).$$

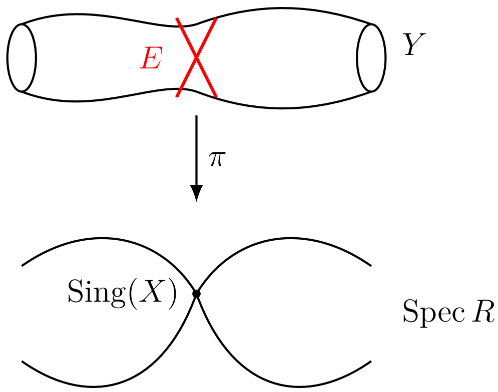

Figure 1: A proper birational morphism resolving the singularity of $\operatorname{Spec} R$. The locus $E \subseteq Y$ is the exceptional set.

Definition 2.2 (Exceptional set)**.** Let $\pi : Y \to X$ be a resolution of singularities. The *exceptional set* is the closed subset

$$E = \operatorname{Exc}(\pi) \subseteq Y$$

where π is not an isomorphism.

For resolutions in characteristic 0, one may arrange several additional properties.

Proposition 2.3 (Standard refinements of resolution)**.** *Let R be as above and let $X = \operatorname{Spec} R$. After replacing a resolution by a sufficiently refined one, we may assume the following.*

(1) *The morphism*

$$\pi : Y \to X$$

is an isomorphism over the non-singular locus of X, that is, over $X \setminus \operatorname{Sing}(X)$.

(2) *The exceptional set*

$$E = \operatorname{Exc}(\pi)$$

is a divisor with simple normal crossings. In other words, every irreducible component of E is non-singular, and locally on Y, the divisor E is defined by a partial product of a regular system of parameters.

(3) *More generally, for any ideal $J \subseteq R$, we may arrange that*

$$V(J\mathcal{O}_Y) \cup E$$

has simple normal crossings.

Definition 2.4 (Simple normal crossings)**.** Let Y be a non-singular variety. A divisor $D \subseteq Y$ has *simple normal crossings*, abbreviated SNC, if for every point $y \in Y$, there is a regular system of parameters

$$x_1, \ldots, x_d$$

in the regular local ring $\mathcal{O}_{Y,y}$ such that D is locally defined by an equation of the form

$$x_1 x_2 \cdots x_r = 0$$

for some $0 \leq r \leq d$.

Thus, locally at a point $y \in Y$, one can write

$$\mathcal{O}_{Y,y} \cong (R', \mathfrak{m}')$$

where

$$\mathfrak{m}' = (x_1, \ldots, x_d),$$

and the exceptional divisor has normal-crossing form, for example

$$E = V(x_1 x_2 \cdots x_r).$$

Remark 2.5. The condition that E be simple normal crossings is crucial because it reduces many local computations on the resolution to computations with monomials in a regular local ring. This is why resolution of singularities is often paired with log-resolution statements: after further blowups, the exceptional divisor and the total transform of a chosen closed subscheme can be made simultaneously monomial in suitable local coordinates.

Example: Blowing Up a Fermat Surface Singularity

Example 2.6. Let

$$R = \frac{k[x,y,z]}{(x^4 + y^4 + z^4)}$$

and let

$$\mathfrak{m} = (x, y, z) \subseteq R.$$

We blow up $X = \operatorname{Spec} R$ at the maximal ideal $\mathfrak{m}$. The blowup is

$$Y = \operatorname{Proj}\left(R \oplus \mathfrak{m}t \oplus \mathfrak{m}^2 t^2 \oplus \cdots\right).$$

We analyze Y on the standard affine charts.

The x-chart

On the chart where x is the chosen generator of the blowup, introduce coordinates

$$u = \frac{y}{x}, \qquad v = \frac{z}{x}.$$

The corresponding affine chart of the blowup is described by

$$k[x,u,v] \Big/ (x^4 + x^4u^4 + x^4v^4).$$

Equivalently,

$$k[x,u,v] \Big/ (x^4(1+u^4+v^4)).$$

The strict transform is obtained by removing the exceptional factor x^4. Hence the strict transform on this chart has coordinate ring

$$k[x,u,v] \Big/ (1+u^4+v^4),$$

that is,

$$k\left[x, \frac{y}{x}, \frac{z}{x}\right] \Big/ \left(1 + \left(\frac{y}{x}\right)^4 + \left(\frac{z}{x}\right)^4\right).$$

Let

$$F_1 = 1 + u^4 + v^4 = 1 + \left(\frac{y}{x}\right)^4 + \left(\frac{z}{x}\right)^4.$$

The exceptional divisor in this chart is

$$E = V(x).$$

To check non-singularity of this chart, compute

$$\frac{\partial F_1}{\partial u} = 4u^3, \qquad \frac{\partial F_1}{\partial v} = 4v^3, \qquad \frac{\partial F_1}{\partial x} = 0.$$

Since $\mathrm{char}(k) = 0$, the equations

$$F_1 = \frac{\partial F_1}{\partial u} = \frac{\partial F_1}{\partial v} = 0$$

would force $u = v = 0$, but then $F_1 = 1 \neq 0$. Hence by Jacobian Criteria there are no singular points on this chart:

$$V\left(F_1, \frac{\partial F_1}{\partial u}, \frac{\partial F_1}{\partial v}\right) = \varnothing.$$

The y-chart

On the chart where y is the chosen generator, set

$$a = \frac{x}{y}, \qquad b = \frac{z}{y}.$$

The strict transform is given by

$$k[a, y, b] \Big/ (a^4 + 1 + b^4),$$

that is,

$$k\left[\frac{x}{y}, y, \frac{z}{y}\right] \Big/ \left(\left(\frac{x}{y}\right)^4 + 1 + \left(\frac{z}{y}\right)^4\right).$$

The same Jacobian computation shows that this chart is non-singular.

The z-chart

On the chart where z is the chosen generator, set

$$c = \frac{x}{z}, \qquad d = \frac{y}{z}.$$

The strict transform is given by

$$k[c, d, z] \Big/ (c^4 + d^4 + 1),$$

that is,

$$k\left[\frac{x}{z}, \frac{y}{z}, z\right] \Big/ \left(\left(\frac{x}{z}\right)^4 + \left(\frac{y}{z}\right)^4 + 1\right).$$

Again, by symmetry, this chart is non-singular. Therefore, the blowup at $\mathfrak{m}$ resolves the singularity in this example.

Remark 2.7. The displayed chart rings above describe the strict transform. The total transform contains a common exceptional factor, such as x^4 on the x-chart. The strict transform is the geometrically relevant component for the resolution, obtained by dividing out this exceptional factor.

Cohomology and Rational Singularities

The goal is to relate R, its normalization, and related extensions such as $R^{1/p}$ or R^{1/p^e} in positive characteristic, to the geometry of a resolution

$$\pi : Y \to \operatorname{Spec} R.$$

A first approximation is to take global sections:

$$\Gamma(Y, \mathcal{O}_Y).$$

For a proper birational morphism from a normal variety Y, one obtains

$$\Gamma(Y, \mathcal{O}_Y) = R^N,$$

where R^N denotes the normalization of R inside its fraction field. However, $\Gamma(Y, \mathcal{O}_Y)$ only records degree-zero cohomological information. Instead, one studies the derived global sections

$$R\Gamma(Y, \mathcal{O}_Y).$$

This is a complex whose cohomology groups are

$$H^i(Y, \mathcal{O}_Y) \qquad (i \geq 0).$$

It can be viewed as a differential graded algebra, and in the setting of rational singularities it is independent, up to quasi-isomorphism, of the chosen resolution of singularities.

Remark 2.8. The derived object $R\Gamma(Y, \mathcal{O}_Y)$ is better behaved than ordinary global sections because it remembers all higher cohomology. Rational singularities are precisely those singularities for which this higher cohomological information contributes nothing beyond the original ring R.

Computing Derived Global Sections by a Čech Complex

Let

$$Y = \bigcup_i U_i$$

be an affine open cover. Then $R\Gamma(Y, \mathcal{O}_Y)$ is represented by the Čech complex

$$\check{C}^\bullet(\{U_i\}, \mathcal{O}_Y):$$

$$\prod_i \Gamma(U_i, \mathcal{O}_Y) \longrightarrow \prod_{i<j} \Gamma(U_i \cap U_j, \mathcal{O}_Y) \longrightarrow \prod_{i<j<\ell} \Gamma(U_i \cap U_j \cap U_\ell, \mathcal{O}_Y) \longrightarrow \cdots.$$

Thus there is a quasi-isomorphism

$$R\Gamma(Y, \mathcal{O}_Y) \simeq \check{C}^\bullet(\{U_i\}, \mathcal{O}_Y).$$

For the previous blowup example, the first terms of the Čech complex are of the form

$$\left(k\left[x, \frac{y}{x}, \frac{z}{x}\right] \Big/ (\cdots)\right) \oplus \left(k\left[\frac{x}{y}, y, \frac{z}{y}\right] \Big/ (\cdots)\right) \oplus \left(k\left[\frac{x}{z}, \frac{y}{z}, z\right] \Big/ (\cdots)\right) \longrightarrow \prod_{i<j} \Gamma(U_i \cap U_j, \mathcal{O}_Y) \longrightarrow \cdots.$$

For example, one of the pairwise intersections is represented by a ring of the form

$$k\left[x, \frac{y}{x}, \frac{z}{x}, \frac{x}{y}\right] \Big/ (\cdots),$$

where the omitted ideal is obtained by imposing the strict-transform equation and the compatibility relations among the displayed coordinates.

Definition 2.9 (Rational singularities)**.** Let R be a domain essentially of finite type over a field of characteristic 0, and let

$$\pi : Y \to X = \operatorname{Spec} R$$

be a resolution of singularities. We say that R, or X, has *rational singularities* if the natural map

$$R \longrightarrow R\Gamma(Y, \mathcal{O}_Y)$$

is a quasi-isomorphism.

Equivalently, R has rational singularities if

$$R \xrightarrow{\sim} \Gamma(Y, \mathcal{O}_Y) = R^N$$

and

$$H^i(Y, \mathcal{O}_Y) = 0 \qquad \text{for all } i > 0.$$

Example 2.10. One can prove that

$$R = \frac{k[x, y, z]}{(x^4 + y^4 + z^4)}$$

does not have rational singularities by showing that, for the resolution Y obtained above,

$$H^1(Y, \mathcal{O}_Y) \neq 0.$$

Open Complements and Local Cohomology

Let $J \subseteq R$ be an ideal generated by

$$J = (f_1, \ldots, f_n).$$

Set

$$U = \operatorname{Spec} R \setminus V(J).$$

The open subset U is covered by the standard affine opens

$$U_i = D(f_i) = \operatorname{Spec} R[f_i^{-1}].$$

The derived global sections $R\Gamma(U, \mathcal{O}_U)$ are computed by the corresponding Čech complex:

$$R \longrightarrow \prod_i R[f_i^{-1}] \longrightarrow \prod_{i<j} R[(f_i f_j)^{-1}] \longrightarrow \cdots .$$

In the context of a resolution Y, one similarly obtains complexes such as

$$0 \longrightarrow R \longrightarrow R\Gamma(Y, \mathcal{O}_Y)[f_1^{-1}] \longrightarrow \cdots ,$$

which represent Čech-type constructions associated to local cohomology. In particular, the local cohomology complex $R\Gamma_J(R)$ is modeled by the cone of

$$R \longrightarrow R\Gamma(U, \mathcal{O}_U),$$

up to the standard cohomological shift conventions.

Remark 2.11. The relationship between Čech complexes and local cohomology is as follows. If $J = (f_1, \ldots, f_n)$, then the ordinary Čech complex on $f_1, \ldots, f_n$ computes $R\Gamma_J(R)$. Equivalently, the complement $\operatorname{Spec} R \setminus V(J)$ is covered by the opens $D(f_i)$, and local cohomology measures the difference between global sections on $\operatorname{Spec} R$ and derived global sections on this open complement.

Why Are Rational Singularities Good?

Normality

Proposition 2.12. *Rational singularities are normal.*

Proof. If R has rational singularities, then by definition the map

$$R \longrightarrow R\Gamma(Y, \mathcal{O}_Y)$$

is a quasi-isomorphism. Taking zeroth cohomology gives an isomorphism

$$R \xrightarrow{\sim} H^0(Y, \mathcal{O}_Y) = \Gamma(Y, \mathcal{O}_Y).$$

For a resolution $Y \to \operatorname{Spec} R$, the ring of global sections is the normalization:

$$\Gamma(Y, \mathcal{O}_Y) \cong R^N.$$

Therefore

$$R \cong R^N,$$

so R is normal. $\square$

Cohen–Macaulayness

Proposition 2.13. *Rational singularities are Cohen–Macaulay.*

Proof. Let $(R, \mathfrak{m})$ be local of dimension d, and suppose R has rational singularities. Then

$$R \simeq R\Gamma(Y, \mathcal{O}_Y)$$

in the derived category. Hence local cohomology gives

$$H^i_{\mathfrak{m}}(R) \cong H^i_{\mathfrak{m}}\Big(R\Gamma(Y, \mathcal{O}_Y)\Big).$$

Using local duality together with the vanishing theorems available for resolutions in characteristic 0, one obtains

$$H^i_{\mathfrak{m}}(R) = 0 \qquad \text{for all } i < \dim R.$$

Therefore

$$\operatorname{depth} R = \dim R,$$

so R is Cohen–Macaulay. $\square$

Remark 2.14. The proof that rational singularities are Cohen–Macaulay is not merely a formal consequence of the definition. It uses deep vanishing results, such as Grauert–Riemenschneider vanishing, together with local duality. The guiding principle is that the higher direct images of $\mathcal{O}_Y$ vanish for rational singularities, and the dualizing sheaf on the smooth resolution has strong vanishing properties.

Grauert–Riemenschneider Vanishing

Theorem 2.15 (Grauert–Riemenschneider vanishing)**.** *Let*

$$\pi : Y \to \operatorname{Spec} R$$

be a resolution of singularities, and let ω_Y be the canonical sheaf of the non-singular variety Y. Then

$$R\Gamma(Y, \omega_Y)$$

is a Cohen–Macaulay complex. Consequently,

$$H^i(Y, \omega_Y) = 0$$

in the relevant cohomological range below the dimension of R.

More precisely, the Cohen–Macaulayness of the complex means that its local cohomology is concentrated in the expected top degree. In the local case, this is the form compatible with local duality and the Cohen–Macaulayness of rings with rational singularities.

Splitting Characterization of Rational Singularities

Theorem 2.16 (Kovács–Bhatt splitting criterion)**.** *Let*

$$X = \operatorname{Spec} R$$

be a variety, or more generally a scheme in the appropriate setting. Then X has rational singularities if and only if for all projective surjective morphisms

$$\pi : Y \to X,$$

the natural map

$$\mathcal{O}_X \longrightarrow R\pi_*\mathcal{O}_Y$$

splits in the derived category.

The lecture diagram can be rendered schematically as follows:

$$\begin{tikzcd}
\text{RoS} \arrow[dr, "\text{projective}"] & Y \arrow[d] & Z \arrow[l, "\text{finite}"'] \arrow[d] \\
 & \operatorname{Spec} R & Z' \arrow[l]
\end{tikzcd}$$

Remark 2.17 (Added Context)**.** The splitting condition says that $\mathcal{O}_X$ is a direct summand of $R\pi_*\mathcal{O}_Y$ in the derived category. This is stronger than merely asking for a splitting on ordinary global sections. The derived splitting controls all higher cohomological obstructions at once.

Reduction to Positive Characteristic

Theorem 2.18 (Smith–Hara–Mehta–Srinivas)**.** *Let $R_{\mathbb{Z}}$ be a ring of finite type over $\mathbb{Z}$, and set*

$$R_{\mathbb{Q}} = R_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Assume that $R_{\mathbb{Q}}$ is Gorenstein. Then $R_{\mathbb{Q}}$ has rational singularities if and only if, for all sufficiently large primes p, the characteristic p reduction

$$R_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{F}_p$$

has strongly F-regular singularities at each localization.

Remark 2.19. This theorem is one of the central bridges between characteristic 0 singularity theory and F-singularity theory. Rational singularities in characteristic 0 correspond, after reduction modulo $p \gg 0$, to strong F-regularity in characteristic p, at least under the stated Gorenstein hypothesis.

3. Lecture–3 May 20, 2026

Let $(A, \mathfrak{m})$ be an f-finite regular local ring, and let

$$0 \neq f \in \mathfrak{m}.$$

The goal is to measure the singularities of the hypersurface

$$R = A/(f).$$

for each integer $e \geq 1$, consider the finite A-algebra A^{1/p^e}. An element of A^{1/p^e} is formally written as a p^e-th root of an element of A. The basic maps under consideration are

$$A \longrightarrow A^{1/p^e}, \qquad 1 \longmapsto f^{a/p^e}.$$

Equivalently, one studies the A-linear map determined by multiplication by f^{a/p^e} after applying the e-fold frobenius-root extension.

Definition 3.1 (f-pure threshold)**.** With notation as above, the *f-pure threshold* of f is

$$\operatorname{fpt}(f) = \sup\left\{ \frac{a}{p^e} \;\middle|\; A \xrightarrow{1 \mapsto f^{a/p^e}} A^{1/p^e} \text{ splits as an } A\text{-module map} \right\}.$$

Remark 3.2. Here splits" means that there exists an A-linear map

$$\psi : A^{1/p^e} \longrightarrow A$$

such that

$$\psi(f^{a/p^e}) = 1.$$

Thus the condition is equivalent to saying that f^{a/p^e} generates a direct summand copy of A inside A^{1/p^e}.

Remark 3.3 (Well-definedness)**.** Nothing pathological occurs when the same rational number is written with different p-power denominators. for example,

$$\frac{a}{p^e} = \frac{ap}{p^{e+1}}.$$

The compatibility is encoded in the following commutative diagram:

$$\begin{tikzcd} A \arrow[r, "1 \mapsto f^{a/p^e}"] \arrow[d, equal] & A^{1/p^e} \arrow[r, "\text{splitting}"] \arrow[d, hook] & A \\ A \arrow[r, "1 \mapsto f^{ap/p^{e+1}}"'] & A^{1/p^{e+1}} \arrow[ur, dashed] & \end{tikzcd}$$

Indeed, the natural inclusion $A^{1/p^e} \hookrightarrow A^{1/p^{e+1}}$ sends

$$f^{a/p^e} \longmapsto f^{ap/p^{e+1}}.$$

Therefore the splitting condition is stable under replacing a/p^e by ap^n/p^{e+n}.

Fact 3.4. One has

$$\boxed{0 < \operatorname{fpt}(f) \leq 1.}$$

Roughly speaking, a larger value of $\operatorname{fpt}(f)$ means that the hypersurface $A/(f)$ is less singular.

Remark 3.5. The upper bound $\operatorname{fpt}(f) \leq 1$ is special to the principal ideal case. for a hypersurface, the threshold reaches its maximal value precisely when the hypersurface has the expected f-singularity property, namely f-purity.

Example 3.6. The following examples are standard benchmarks:

$$\operatorname{fpt}\Big(xy \in \mathbb{F}_p[x,y]\Big) = 1,$$
$$\operatorname{fpt}\Big(x^2 - yz \in k[x,y,z]\Big) = 1,$$

and, for the cusp,

$$\operatorname{fpt}\Big(y^2 - x^3 \in k[x,y]\Big) = \begin{cases} \dfrac{5}{6}, & \text{if } p \equiv 1 \pmod 6, \\ \dfrac{1}{2}, & \text{if } p = 2, \\ \dfrac{2}{3}, & \text{if } p = 3, \\ \dfrac{5}{6} - \dfrac{1}{6p}, & \text{if } p \equiv 5 \pmod 6. \end{cases}$$

The notes also record the slogan

$$\lim_{p\to\infty} \operatorname{fpt}(f) = 1 \qquad \text{if and only if} \qquad \operatorname{fpt}(f) = 1.$$

Lemma 3.7. *Let $R = A/(f)$. Then R is f-split if and only if*

$$\operatorname{fpt}(f) = 1.$$

Proof. We prove both directions carefully. first suppose that $\operatorname{fpt}(f) = 1$. Since $\operatorname{fpt}(f) \leq 1$, this means that the splitting condition holds arbitrarily close to 1. In particular, the map

$$A \longrightarrow A^{1/p}, \qquad 1 \longmapsto f^{(p-1)/p}$$

splits. Hence there exists an A-linear map

$$\psi : A^{1/p} \longrightarrow A$$

such that

$$\psi\Big(f^{(p-1)/p}\Big) = 1.$$

Consider the composite

$$\begin{tikzcd} A \arrow[r, "\text{frobenius-root map}"] \arrow[d, equal] & A^{1/p} \arrow[r, "\cdot f^{(p-1)/p}"] & A^{1/p} \arrow[r, "\psi"] & A \\ 1 \arrow[r, mapsto] & 1^{1/p} \arrow[r, mapsto] & f^{(p-1)/p} \arrow[r, mapsto] & 1. \end{tikzcd}$$

Let

$$\Phi = \psi \circ \Big(\cdot f^{(p-1)/p}\Big) : A^{1/p} \longrightarrow A.$$

Then Φ is A-linear and satisfies

$$\Phi(1^{1/p}) = 1.$$

Moreover,

$$\begin{aligned} \Phi\Big((f^{1/p})A^{1/p}\Big) &= \psi\Big(f^{(p-1)/p}(f^{1/p})A^{1/p}\Big) \\ &= \psi\Big(fA^{1/p}\Big) \\ &\subseteq f\psi(A^{1/p}) \\ &\subseteq (f). \end{aligned}$$

Therefore Φ descends modulo (f) to an R-linear map

$$\overline{\Phi} : R^{1/p} \longrightarrow R.$$

The descent is represented by the commutative diagram

$$\begin{tikzcd}
(f^{1/p})A^{1/p} \arrow[r] \arrow[d, hook] & (f) \arrow[d, hook] \\
A^{1/p} \arrow[r, "\Phi"] \arrow[d] & A \arrow[d] \\
A^{1/p}/(f^{1/p})A^{1/p} \arrow[r, "\overline{\Phi}"] \arrow[d] & A/(f) \arrow[d] \\
0 & 0.
\end{tikzcd}$$

Since

$$A^{1/p}/(f^{1/p})A^{1/p} \cong R^{1/p}$$

and

$$\overline{\Phi}(1_R^{1/p}) = 1_R,$$

the frobenius map

$$R \longrightarrow R^{1/p}$$

splits. Thus $R = A/(f)$ is f-split. Conversely, suppose that $R = A/(f)$ is f-split. Then there exists an R-linear map

$$\varphi : R^{1/p} \longrightarrow R$$

such that

$$\varphi(1_R^{1/p}) = 1_R.$$

By the usual fedder-type lifting criterion for a hypersurface in a regular local ring, this is equivalent to the existence of an A-linear map

$$\psi : A^{1/p} \longrightarrow A$$

such that

$$\psi\Big(f^{(p-1)/p}\Big) = 1.$$

Hence the map

$$A \longrightarrow A^{1/p}, \qquad 1 \longmapsto f^{(p-1)/p}$$

splits. Therefore

$$\frac{p-1}{p} \leq \mathrm{fpt}(f).$$

Iterating the same argument for the e-fold frobenius gives splittings

$$A \longrightarrow A^{1/p^e}, \qquad 1 \longmapsto f^{(p^e-1)/p^e}$$

for all $e \geq 1$. Consequently
$$\frac{p^e - 1}{p^e} \leq \operatorname{fpt}(f) \qquad \text{for all } e \geq 1.$$
Letting $e \to \infty$, we obtain
$$1 \leq \operatorname{fpt}(f).$$
Since always $\operatorname{fpt}(f) \leq 1$, it follows that
$$\operatorname{fpt}(f) = 1.$$
$\square$

The perfect Closure and the Absolute Integral Closure

Remark 3.8 (Rodriguez–Villalobos)**.** Define
$$A_{\text{perf}} = \bigcup_{e>0} A^{1/p^e}.$$
This ring is usually not Noetherian. for example, if
$$A = \mathbb{F}_p[[x]],$$
then the ideal
$$(x, x^{1/p}, x^{1/p^2}, \ldots) \subseteq A_{\text{perf}}$$
is not finitely generated. Using A_{perf}, the f-pure threshold can be rewritten as
$$\operatorname{fpt}(f) = \sup \left\{ \frac{a}{p^e} \;\middle|\; A \xrightarrow{1 \mapsto f^{a/p^e}} A_{\text{perf}} \text{ is pure} \right\}.$$
If $A = \widehat{A}$, then purity can be detected by splitting in this setting.

Now set
$$A^+ = \text{the integral closure of } A \text{ in } \overline{K(A)}.$$
Equivalently,
$$A^+ = \bigcup_{\substack{A \subseteq S \subseteq \overline{K(A)} \\ S \text{ finite over } A}} S.$$
The ring A^+ is generally non-Noetherian. In these terms, one can write
$$\operatorname{fpt}(f) = \sup \left\{ t \in \mathbb{Q} \;\middle|\; A \xrightarrow{1 \mapsto f^t} A^+ \text{ is pure} \right\},$$
with the additional note that if $A = \widehat{A}$, then the relevant pure maps split.

Remark 3.9. The notation f^t for rational t should be understood after passing to a finite extension S of A in which the required root of f exists. Thus the expression
$$A \xrightarrow{1 \mapsto f^t} A^+$$
packages all finite extensions containing the relevant fractional power of f.

Characteristic 0

Let $(A, \mathfrak{m})$ be a regular local ring essentially of finite type over a field k, where

$$\operatorname{char} k = 0,$$

and let

$$0 \neq f \in \mathfrak{m}.$$

The characteristic-zero analogue of the preceding positive-characteristic splitting condition is recorded as

$$\sup \left\{ t \in \mathbb{Q} \;\middle|\; \forall S \supseteq A \text{ finite with } f^t \in S,\ A \xrightarrow{1 \mapsto f^t} S \text{ splits} \right\}.$$

Definition 3.10 (log canonical threshold)**.** The *log canonical threshold* of f is

$$\operatorname{lct}(f) = \sup \left\{ t \in \mathbb{Q}_{\geq 0} \;\middle|\; \begin{array}{l} \forall S \supseteq A \text{ finite with } f^t \in S, \\ \text{for every resolution of singularities } Y \to \operatorname{Spec} S, \\ A \to S \xrightarrow{1 \mapsto f^t} R\Gamma(Y, \mathcal{O}_Y) \text{ splits} \end{array} \right\}.$$

Remark 3.11. Here RoS" in the lecture notes means resolution of singularities." The object

$$R\Gamma(Y, \mathcal{O}_Y)$$

is the derived global section complex of the structure sheaf. The splitting condition should be interpreted in the derived category, or equivalently after applying the appropriate derived Hom functor. In the hypersurface case, this threshold agrees with the usual birational-geometric log canonical threshold.

Example 3.12. The characteristic-zero analogues of the previous examples are

$$\begin{aligned} \operatorname{lct}\Big(xy \in \mathbb{Q}[x, y]\Big) &= 1, \\ \operatorname{lct}\Big(x^2 - yz \in \mathbb{Q}[x, y, z]\Big) &= 1, \\ \operatorname{lct}\Big(y^2 - x^3 \in \mathbb{C}[x, y]\Big) &= \frac{5}{6}. \end{aligned}$$

Fact 3.13. for $0 \neq f \in \mathfrak{m}$, one has:

(1)
$$0 < \operatorname{lct}(f) \leq 1.$$

(2)
$$\operatorname{lct}(f) = 1 \quad \Longleftrightarrow \quad A/(f) \text{ is log canonical.}$$

Moreover, in the Gorenstein setting, rational singularities imply log canonical singularities:

$$A/(f) \text{ Gorenstein rational} \quad \Longrightarrow \quad A/(f) \text{ log canonical.}$$

Theorem 3.14 (Takagi–Watanabe; Smith; Hara; Mehta–Srinivas)**.** *Let*

$$A_{\mathbb{Z}} = \mathbb{Z}[x_1, \ldots, x_n],$$

let

$$A_{\mathbb{Q}} = A_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q},$$

and, for a prime p, let

$$A_p = A_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{F}_p$$

Let

$$0 \neq f \in (x_1, \ldots, x_n)A_{\mathbb{Z}},$$

and let

$$f_p = \mathrm{im}(f) \in A_p$$

be its reduction modulo p. Then

$$\lim_{p\to\infty} \mathrm{fpt}(f_p \in A_p) = \mathrm{lct}(f \in A_{\mathbb{Q}}).$$

Remark 3.15. This theorem expresses the fundamental compatibility between f-singularities in large positive characteristic and birational singularities in characteristic zero. The log canonical threshold is recovered as the limiting value of f-pure thresholds of reductions modulo p.

Mixed Characteristic

Let $(A, \mathfrak{m})$ be a complete regular local ring of mixed characteristic $(0, p > 0)$. Thus

$$\operatorname{char} A = 0 \qquad \text{and} \qquad \operatorname{char}(A/\mathfrak{m}) = p > 0.$$

Example 3.16. Typical examples include

$$A = \mathbb{Z}_p[[x_1, \ldots, x_n]], \qquad \mathfrak{m} = (p, x_1, \ldots, x_n),$$

and

$$A = \mathbb{Z}_p[[x_2, \ldots, x_n]]\left[p^{1/p}\right], \qquad \mathfrak{m} = (p^{1/p}, x_2, \ldots, x_n).$$

Theorem 3.17 (Bhatt)**.** *Let A be as above. Then the p-adic completion*

$$(A^+)^{\wedge p}$$

of A^+ is a big Cohen–Macaulay A-algebra. Equivalently, one may use the $\mathfrak{m}$-adic completion

$$\widehat{A^+}^{\mathfrak{m}}.$$

Remark 3.18 (Added Context)**.** A big Cohen–Macaulay A-algebra B is an A-algebra on which every system of parameters of A is a regular sequence. Bhatt's theorem is one of the key inputs allowing one to formulate mixed-characteristic analogues of characteristic-p splitting thresholds.

Definition 3.19 (plus pure threshold)**.** Let $0 \neq f \in \mathfrak{m}$. The *plus pure threshold* of f is

$$\begin{aligned}\operatorname{ppt}(f) &= \sup\left\{t \in \mathbb{Q}_{>0} \;\middle|\; A \xrightarrow{1 \mapsto f^t} (A^+)^{\wedge p} \text{ is pure, equivalently splits}\right\}\\ &= \sup\left\{t \in \mathbb{Q}_{>0} \;\middle|\; f^t \notin \mathfrak{m}(A^+)^{\wedge p}\right\} \qquad \text{since } (A^+)^{\wedge p} \text{ is faithfully flat over } A^+\\ &= \sup\left\{t \in \mathbb{Q}_{>0} \;\middle|\; f^t \notin \mathfrak{m}A^+\right\}\\ &= \sup\left\{t \in \mathbb{Q}_{>0} \;\middle|\; f^t \notin \mathfrak{m}S \quad \forall S \supseteq A \text{ finite with } f^t \in S\right\}.\end{aligned}$$

Remark 3.20. The equality between the splitting formulation and the condition

$$f^t \notin \mathfrak{m}(A^+)^{\wedge p}$$

is a mixed-characteristic analogue of the elementary local criterion for a map

$$A \to B, \qquad 1 \mapsto b,$$

to split when B behaves like a sufficiently large Cohen–Macaulay algebra. In a local situation, the obstruction to splitting is measured by whether b lands inside the extension of the maximal ideal.

Fact 3.21. for $0 \neq f \in A$, the thresholds satisfy

$$\operatorname{fpt}(\overline{f} \in A/p) \leq \operatorname{ppt}(f \in A) \leq \operatorname{lct}(f \in A_{\mathbb{Q}}).$$

Example 3.22. The notes record the following mixed-characteristic comparison:

$$\operatorname{ppt}\Big(p^2 - x^3 \in A \text{ of mixed characteristic } (0,p)\Big) = \operatorname{fpt}\Big(y^2 - x^3 \in \mathbb{F}_p[x,y]\Big).$$

They also record the expression

$$\begin{aligned}\operatorname{ppt}(8 + y^3 + z^3) &\leq 1,\\ &< \frac{3}{4}.\end{aligned}$$

Question 3.23 (Tucker's Bane)**.** Compute

$$\operatorname{ppt}\Big(x^3 + 27 \in \mathbb{Z}_3[x]\Big).$$

4. Lecture–4 May 21, 2026

Definition 4.1 (Big Cohen–Macaulay algebras)**.** Let $(R, \mathfrak{m})$ be an excellent local domain. In the present notes, "excellent" means, for example, complete or essentially of finite type over a field k, over $\mathbb{Z}$, over $\mathbb{Z}_{(p)}$, and similar standard excellent base rings.

An R-algebra B is called **weakly balanced big Cohen–Macaulay** if every system of parameters on R is a regular sequence on B. It is called **balanced big Cohen–Macaulay** if, in addition,

$$\mathfrak{m}B \neq B.$$

We abbreviate this by saying that B is a **BCM algebra**.

Remark 4.2 (Added Context)**.** The adjective "balanced" means that the regular sequence condition is required for every system of parameters, not merely for one chosen system of parameters. The condition $\mathfrak{m}B \neq B$ rules out the degenerate case in which the parameter sequence is vacuously regular because the algebra has collapsed.

Theorem 4.3 (Hochster–Huneke, André, Gabber)**.** *Big Cohen–Macaulay algebras exist.*

Theorem 4.4. *Let $(R, \mathfrak{m})$ be an excellent regular local ring, and let B be a BCM R-algebra. Then B is flat as an R-module.*

Remark 4.5. For a regular local ring, every system of parameters is a regular system of parameters. Thus the BCM condition forces the homological behavior of B over R to be as good as possible. The theorem is a big Cohen–Macaulay analogue of the local criterion for flatness.

BCM-Regularity Along a Fixed BCM Algebra

Definition 4.6 (Weak BCM-regularity along a BCM algebra)**.** Let B be a BCM R-algebra. We say that R is **weakly BCM-regular along** B if the structure map

$$R \longrightarrow B$$

is pure. If R is complete, this is equivalent to saying that $R \to B$ splits as a map of R-modules.

Remark 4.7. A map $R \to B$ is pure if, for every R-module M, the induced map

$$M \longrightarrow M \otimes_R B$$

is injective. When R is complete local, purity of certain module-finite or sufficiently controlled maps can be tested by splitting criteria; this is the sense in which purity and splitting are being identified in the lecture notes.

Theorem 4.8 (Deformation statement for BCM-regularity)**.** *Let $(R, \mathfrak{m})$ be an excellent local domain. Assume that R is Gorenstein, and let $0 \neq f \in \mathfrak{m}$. Let B be a BCM R-algebra. Suppose that $R/(f)$ is BCM-regular along $B/(f)$. Then R is BCM-regular. In the notation below, put* $\dim R = d$*.*

Proof. The proof is phrased in terms of local cohomology. It suffices to show the required injectivity in the following diagram. Let

$$\eta \in H^d_{\mathfrak{m}}(R).$$

Without loss of generality, assume that

$$f\eta = 0.$$

The relevant portion of the long exact sequence in local cohomology associated to

$$0 \longrightarrow R \xrightarrow{f} R \longrightarrow R/(f) \longrightarrow 0$$

is
$$H^{d-1}_{\mathfrak{m}}(R/(f)) \xrightarrow{\gamma} H^d_{\mathfrak{m}}(R) \xrightarrow{f} H^d_{\mathfrak{m}}(R).$$
The corresponding sequence for B and B/fB fits into the commutative diagram

$$\begin{tikzcd}
H^{d-1}_{\mathfrak{m}}(R) \arrow[r] & H^{d-1}_{\mathfrak{m}}(R/(f)) \arrow[r, "\gamma"] \arrow[d] & H^d_{\mathfrak{m}}(R) \arrow[r, "f"] \arrow[d] & H^d_{\mathfrak{m}}(R) \arrow[d] \\
0 \arrow[r] & H^{d-1}_{\mathfrak{m}}(B/fB) \arrow[r] & H^d_{\mathfrak{m}}(B) \arrow[r, "f"] & H^d_{\mathfrak{m}}(B).
\end{tikzcd}$$

The lower-left zero reflects the Cohen–Macaulay behavior forced by the BCM condition. Thus, to prove that the image of η in $H^d_{\mathfrak{m}}(B)$ is nonzero whenever η is nonzero, it is enough to prove injectivity at the indicated stage.

Since $f\eta = 0$, exactness gives an element
$$\xi \in H^{d-1}_{\mathfrak{m}}(R/(f))$$
such that
$$\gamma(\xi) = \eta.$$
The hypothesis that $R/(f)$ is BCM-regular along $B/(f)$ gives the required injectivity after mapping to $H^{d-1}_{\mathfrak{m}}(B/fB)$. Therefore the image of ξ does not vanish unless ξ already vanishes in the relevant quotient. By commutativity of the diagram, this forces the image of η in $H^d_{\mathfrak{m}}(B)$ to be nonzero.

It remains to explain why this injectivity criterion is enough. Because R is Gorenstein, the top local cohomology module $H^d_{\mathfrak{m}}(R)$ is the injective hull of the residue field. Consequently, purity or splitting of $R \to B$ can be tested on the induced map on top local cohomology. This proves BCM-regularity of R. $\square$

Remark 4.9. The displayed diagram is obtained from the long exact local cohomology sequences associated to multiplication by f on R and on B. The vertical maps are induced by the structural map $R \to B$.

Remark 4.10. If R is Gorenstein in characteristic $p > 0$, then BCM-regularity for all BCM algebras is equivalent to strong F-regularity.

Question 4.11 (Characteristic zero comparison)**.** Let R be Gorenstein of characteristic zero. Is it true that
$$R \text{ is BCM-regular for all BCM algebras} \iff R \text{ has rational singularities?}$$
One implication is known:
$$\text{BCM-regular for all } B \implies \text{rational singularities.}$$

Examples and Sources of BCM Algebras

Characteristic $p > 0$

In characteristic $p > 0$, one important source of BCM algebras is the absolute integral closure R^+. The perfect closure R_{perf} is almost BCM.

Mixed Characteristic

In mixed characteristic, the following completions of the absolute integral closure are BCM:

$$(R^+)^{\wedge p}, \qquad (R^+)^{\wedge \mathfrak{m}}.$$

Characteristic Zero

In characteristic zero, various constructions of BCM algebras are available. The lecture notes do not specify a single preferred construction here.

Equational Lemmas

Lemma 4.12 (Equational lemma; Huneke–Lyubeznik)**.** *Let R be a domain of characteristic $p > 0$, and let $J \subseteq R$ be an ideal. Let*

$$\eta \in H^i_J(R).$$

Suppose that the set

$$\{\eta, \eta^p, \eta^{p^2}, \ldots\}$$

generates a finitely generated R-module. Then there exists a finite domain extension

$$R \hookrightarrow S$$

such that the image of η vanishes:

$$\eta \longmapsto 0 \quad \textit{in} \quad H^i_{JS}(S).$$

Remark 4.13. If

$$\eta \in H^i_{\mathfrak{m}}(R^+) \qquad \text{for } i < \dim R,$$

then η is already represented over some finite extension R' of R. If the equational lemma applies, then after passing to a further finite extension one kills η. This is the mechanism behind the vanishing

$$H^i_{\mathfrak{m}}(R^+) = 0 \qquad \text{for } i < \dim R$$

in characteristic $p > 0$.

Lemma 4.14 (Equational lemma II; Hochster–Huneke, Bhatt)**.** *Let*

$$Y \longrightarrow \operatorname{Spec} R$$

be proper, where R is a Noetherian integral domain of characteristic $p > 0$. For every class

$$\eta \in H^i(Y, \mathcal{O}_Y),$$

there exists a finite surjective integral morphism

$$Y' \longrightarrow Y$$

such that

$$\eta \longmapsto 0 \quad \text{in} \quad H^i(Y', \mathcal{O}_{Y'}).$$

Bhatt's Theorem and Derived Splinters

Theorem 4.15 (Bhatt)**.** *Let R be an excellent domain, and let*

$$Y \longrightarrow \operatorname{Spec} R$$

be proper and surjective. There is a factorization in the derived category

$$\begin{tikzcd} R \arrow[r] \arrow[rr, bend right, "1 \mapsto 1"'] & R\Gamma(Y, \mathcal{O}_Y) \arrow[r] & R^+. \end{tikzcd}$$

In particular, after passing to suitable finite covers, one can kill the higher cohomology classes appearing in $R\Gamma(Y, \mathcal{O}_Y)$.

Definition 4.16 (Splinters and derived splinters)**.** Let R be a domain.

(1) R is a **splinter** if every finite extension

$$R \hookrightarrow S$$

splits as a map of R-modules. Equivalently, $R \to R^+$ is pure.

(2) R is a **derived splinter** if, for every proper surjective morphism

$$Y \longrightarrow \operatorname{Spec} R,$$

the natural map

$$R \longrightarrow R\Gamma(Y, \mathcal{O}_Y)$$

splits in the derived category.

(3) R is a **birational derived splinter** if the same splitting condition holds for every blowup

$$Y \longrightarrow \operatorname{Spec} R.$$

Remark 4.17. In characteristic $p > 0$, Bhatt's theorem implies

$$\text{splinter} \quad \Longrightarrow \quad \text{derived splinter.}$$

Remark 4.18. Bhargav Bhatt gave a mixed characteristic version of the equational lemma and deduced that

$$(R^+)^{\wedge p}$$

is BCM. In mixed characteristic, one also has a factorization

$$R \longrightarrow R\Gamma(Y, \mathcal{O}_Y) \longrightarrow (R^+)^{\wedge p}.$$

Implication Diagrams

Characteristic $p > 0$

In characteristic $p > 0$, the relationships discussed in the lecture are summarized by the following diagram:

$$\begin{tikzcd}
\text{strongly } F\text{-regular} \arrow[r, Rightarrow] & \text{BCM-regular for all } B \arrow[r, Rightarrow] & \text{splinter} \arrow[r, Rightarrow, "F\text{-finite}"] \arrow[dl, Rightarrow, dashed, "?"', "\text{known if Gorenstein}"] \arrow[d, equal] \arrow[dr, Rightarrow] & F\text{-split} \\
 & \text{strongly } F\text{-regular} & \text{derived splinter} & \text{normal and CM.}
\end{tikzcd}$$

Remark 4.19 (Open question)**.** Does

$$\text{splinter} \quad \Longrightarrow \quad \text{strongly } F\text{-regular}$$

hold in characteristic $p > 0$? This is known when R is Gorenstein.

The Cohen–Macaulayness mechanism is reflected in the vanishing

$$\boxed{H^i_{\mathfrak{m}}(R) \hookrightarrow H^i_{\mathfrak{m}}(R^+) = 0 \qquad \text{for all } i < \dim R.}$$

Remark 4.20. The injection $H^i_{\mathfrak{m}}(R) \hookrightarrow H^i_{\mathfrak{m}}(R^+)$ comes from purity of $R \to R^+$ when R is a splinter. The vanishing of the right-hand side is the big Cohen–Macaulay property of R^+ in characteristic $p > 0$. Thus splinters in positive characteristic are Cohen–Macaulay.

Characteristic Zero

In characteristic zero, the lecture records the following relationships:

$$
\begin{tikzcd}
\begin{array}{c}\text{BCM-regular}\\ \text{for all } B\end{array} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \text{rational} \arrow[r, equal] \arrow[d, Rightarrow] \arrow[dr, Rightarrow] & \begin{array}{c}\text{derived}\\ \text{splinter}\end{array} \arrow[r, equal] & \begin{array}{c}\text{birational}\\ \text{derived}\\ \text{splinter}\end{array} \\
\text{log-terminal} \arrow[d, Rightarrow] & \text{normal} & \text{splinter} \arrow[u, Rightarrow, "\text{BCM}"'] & \\
\text{log-canonical} & & &
\end{tikzcd}
$$

Conjecture 4.21. In characteristic zero,

$$\text{rational singularities} \quad \Longrightarrow \quad \text{BCM-regular for all BCM algebras.}$$

This is known if R is Gorenstein.

Remark 4.22. The equality between rational singularities and derived splinters in characteristic zero is usually understood through resolution of singularities and Grauert–Riemenschneider-type vanishing. The birational derived splinter condition tests the same splitting only on birational proper maps, such as blowups.

Mixed Characteristic

In mixed characteristic, the lecture records:

$$\text{BCM-regular for all } B \quad \Longrightarrow \quad \text{splinter} = \text{derived splinter.}$$

The corresponding diagram is

$$
\begin{tikzcd}
\text{BCM-regular for all } B \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \text{splinter} \arrow[r, Rightarrow] & \text{normal and Cohen–Macaulay} \\
R \to (R^+)^{\wedge p} \text{ is pure} & &
\end{tikzcd}
$$

Remark 4.23. The mixed characteristic theory is subtler because Frobenius methods are not directly available. Perfectoid methods and the construction of $(R^+)^{\wedge p}$ as a BCM algebra supply the replacement for the positive characteristic argument.